

Now, for the V_1 term, first observe that

$$\begin{aligned}
\text{tr } \Sigma(\Sigma - \kappa u I_d)(\Sigma + \kappa I_d)^{-2} &= \text{tr } \Sigma(\Sigma - \frac{\kappa \text{df}_2}{n - \text{df}_2} I_d)(\Sigma + \kappa I_d)^{-2} \\
&= \text{df}_2 - \frac{\text{df}_2}{n - \text{df}_2} \cdot \kappa \text{tr } \Sigma(\Sigma + \kappa I_d)^{-2} \\
&= \text{df}_2 - \frac{\text{df}_2}{n - \text{df}_2} (\text{df}_1 - \text{df}_2) \\
&= \frac{\text{df}_2}{n - \text{df}_2} (n - \text{df}_1).
\end{aligned}$$

We deduce that

$$V_1 = \sigma^2 \cdot (1 - \text{df}_1/n) \frac{\kappa}{\lambda} \cdot \frac{\text{df}_2}{n - \text{df}_2} = \sigma^2 \cdot \frac{\text{df}_2}{n - \text{df}_2} =: V,$$

where we have used the identity $\kappa - \lambda = \kappa \text{df}_1/n$, which defines κ .

We now handle the B_1 term. First observe that $u + 1 = n/(n - \text{df}_2)$, and so one computes

$$\kappa^2(u + 1) \text{tr } \Gamma \Sigma(\Sigma + \kappa I_d)^{-2} = \kappa^2 \frac{n}{n - \text{df}_2} \text{tr } \Gamma \Sigma(\Sigma + \lambda I_d)^{-2} =: B,$$

which is the classical formula for the bias term.

To finalize, observe that

$$\begin{aligned}
\text{tr } \Delta \Sigma C_{2,1} K^{-2} &= \text{tr } \Delta \Sigma (p_2(1 + p_1 u) \Sigma^2 + u(p_1 \Sigma + \kappa I_d)^2) (\Sigma + \kappa I_d)^{-2} \\
&= p_2(1 + p_1 u) \text{tr } \Delta \Sigma^3 (\Sigma + \kappa I_d)^{-2} + u \text{tr } \Delta \Sigma (p_1 \Sigma + \kappa I_d)^2 (\Sigma + \kappa I_d)^{-2} =: \zeta,
\end{aligned}$$

which concludes the proof. $\square$

E.2 Proof of Corollary 1

Indeed, here we have $\kappa \rightarrow 0$ and $u \rightarrow \phi/(1 - \phi)$ in the limit $\lambda \rightarrow 0^+$. Theorem 1 then gives

$$\begin{aligned}
E_{test}(\hat{f}_{CL}) &\simeq V + B + \zeta, \text{ where } V = \frac{\sigma^2 \phi}{1 - \phi}, \quad B = 0, \\
\zeta &= \frac{p_2 c^2}{1 - \phi} (p_2(1 - \phi + p_1 \phi) + p_1^2 \phi) = \frac{p_2 c^2}{1 - \phi} (p_2(1 - p_2 \phi) + p_1^2 \phi) \\
&= \frac{p_2 c^2}{1 - \phi} (p_2 + (p_1 - p_2) \phi) = p_2^2 c^2 + \frac{p_2 p_1 c^2 \phi}{1 - \phi}.
\end{aligned}$$

For small ϕ , this further gives $E_{test}(\hat{f}_{CL}) \simeq \sigma^2 \phi/(1 - \phi) + p_2^2 c^2 + O(\phi^2) \simeq \sigma^2 d/n + p_2^2 c^2 + O(\phi^2)$. $\square$

E.3 Proof of Corollary 2

The setup can be seen as a special instance of the setup considered in the proof of Proposition 2 (cf. Appendix E.1), since it corresponds to taking $\Sigma_1 = (1 - \alpha)\Sigma/p_1$, and $\Sigma_2 = \alpha\Sigma/p_2$. We must have

$$\frac{1}{e_1} = 1 + \phi \bar{\text{tr}} \Sigma_1 K^{-1} = 1 + \frac{(1 - \alpha)\phi/p_1}{(1 - \alpha)e_1 + \alpha e_2 + \lambda}, \quad (99)$$

$$\frac{1}{e_2} = 1 + \phi \bar{\text{tr}} \Sigma_2 K^{-1} = 1 + \frac{\alpha\phi/p_2}{(1 - \alpha)e_1 + \alpha e_2 + \lambda}. \quad (100)$$

At least for $\lambda = 0$ and $0 < \phi < 1$, these equations can be solved explicitly to get $e_1, e_2 \geq 0$ but the resulting formulae are rather complicated, and therefore are omitted altogether. In any case, theorem 1 correctly predicts the test error, as can be seen in Figure 9.

A particular case where things are solvable to give simple expressions, is when $\phi \rightarrow 0^+$. In this limit, it is easy to see that $e_1 = e_2 = 1$ and $u_1 = u_2 = 0$. This gives

$$K = \Sigma + \lambda I_d, \quad (101)$$

$$L' = \Sigma, \quad (102)$$

$$C_1 = (1 - \alpha)\Sigma, \quad (103)$$

$$C_2 = \alpha\Sigma, \quad (104)$$

$$D_k = \Sigma, \quad (105)$$

$$\lambda^2 r^{(2)}(A, \Sigma) \simeq \lambda^2 \operatorname{tr} A \Sigma (\Sigma + \lambda I_d)^{-2} = \lambda \cdot (\operatorname{tr} A \Sigma (\Sigma + \lambda)^{-1} - \operatorname{tr} A \Sigma^2 (\Sigma + \lambda I_d)^{-2}), \quad (106)$$

$$r_1^{(3)}(A, \Sigma) \simeq p_1 \operatorname{tr} A \Sigma_1 C_1 K^{-2} = p_1^1 \alpha^2 \operatorname{tr} A \Sigma^2 (\Sigma + \lambda I_d)^{-2}, \quad (107)$$

$$r_2^{(3)}(A, \Sigma) \simeq p_2 \operatorname{tr} A \Sigma_2 C_2 K^{-2} = p_2^2 (1 - \alpha)^2 \operatorname{tr} A \Sigma^2 (\Sigma + \lambda I_d)^{-2}, \quad (108)$$

$$r_1^{(4)}(A, \Sigma) \simeq p_1 \operatorname{tr} A \Sigma_1 D_1 K^{-2} = p_1 \alpha \operatorname{tr} A \Sigma^2 (\Sigma + \lambda I_d)^{-2}, \quad (109)$$

$$r_2^{(4)}(A, \Sigma) \simeq p_2 \operatorname{tr} A \Sigma_2 D_2 K^{-2} = p_2 (1 - \alpha) \operatorname{tr} A \Sigma^2 (\Sigma + \lambda I_d)^{-2}. \quad (110)$$

We deduce that

$$V_1 = \sum_{j=1}^2 p_j \frac{\sigma_j^2}{n} r_j^{(4)}(I_d, \Sigma) = ((1 - \alpha)p_1 \sigma_1^2 + \alpha p_2 \sigma_2^2) \frac{\operatorname{df}_2(\lambda; \Sigma)}{n}, \quad (111)$$

$$B_1 = r_2^{(3)}(\Delta, \Sigma) + \lambda^2 r^{(2)}(\Gamma, \Sigma) \xrightarrow{\lambda \rightarrow 0^+} p_2^2 (1 - \alpha)^2 \operatorname{tr} \Delta. \quad (112)$$

Putting things together then gives

$$E_{test}(\hat{f}_{CL}) \simeq B_1 + V_1 \simeq p_2^2 (1 - \alpha)^2 \operatorname{tr} \Delta + (\alpha p_1 \sigma_1^2 + (1 - \alpha) p_2 \sigma_2^2) \frac{d}{n},$$

as claimed. $\square$

E.4 Proof of Corollary 3

Applying the first part of Corollary 1 recursively gives for any iteration $t \geq 1$,

$$E_{test}(\hat{f}_{CL}^{(t)}) \simeq c_t^2 \simeq \bar{E} + p_2^2 c_{t-1}^2 \simeq \dots \simeq p_2^{2t} c_0^2 + \frac{1 - p_2^{2t}}{1 - p_2^2} \bar{E}, \text{ with } \bar{E} := \frac{\sigma^2 \phi}{1 - \phi}.$$

Iterating the above gives

$$c_{t+1}^2 \simeq \frac{\sigma^2 \phi_t}{1 - \phi_t} + p_2^2 c_t^2, \quad \phi_t = d/N_t, \quad N_t = n, \quad c_0^2 = c^2. \quad (113)$$

Setting $\bar{E} := \sigma^2 \phi / (1 - \phi) \simeq \sigma^2 d / n$, we get

$$\begin{aligned} E_{test}(\hat{f}_{CL}^{(t+1)}) &\simeq c_{t+1}^2 \simeq p_2^2 c_t^2 + \frac{\sigma^2 \phi_t}{1 - \phi_t} \\ &\simeq p_2^2 \left(p_2^2 c_{t-1}^2 + \frac{\sigma^2 \phi_{t-1}}{1 - \phi_{t-1}} \right) + \frac{\sigma^2 \phi_t}{1 - \phi_t} \\ &\simeq p_2^{2(1+1)} c_{t-1}^2 + p_2^2 \frac{\sigma^2 \phi_{t-1}}{1 - \phi_{t-1}} + \frac{\sigma^2 \phi_t}{1 - \phi_t} \\ &\vdots \\ &\simeq p_2^{2(t+1)} c_0^2 + \sum_{0 \leq j \leq t} \frac{\sigma^2 \phi_j}{1 - \phi_j} p_2^{2(t-j)} \\ &= p_2^{2(t+1)} c^2 + \bar{E} \sum_{0 \leq j \leq t} p_2^{2j} \\ &= p_2^{2(t+1)} c^2 + \frac{1 - p_2^{2(t+1)}}{1 - p_2^2} \bar{E}. \end{aligned}$$

In particular, we if p_2 is bounded away from 1 (i.e if $p_1 := 1 - p_2 = \Omega(1)$), we get

$$E_{test}(\hat{f}_{CL}^{(t)}) \simeq \frac{1}{1 - p_2^2} \bar{E} + p_2^{2t} c^2,$$

for large t . The first part is just a constant multiple of the scaling law we would have with training on a dataset comprising of n units of clean data.

On the other hand, we have

$$\lim_{p_2 \rightarrow 1} E_{test}(\hat{f}_{CL}^{(t)}) \simeq c^2 + t\bar{E}.$$

This is an increasing function of t , lower-bounded by $c^2 + \bar{E}$. We recover the classical picture, in which model collapse prevails (depending on the size of c^2 , as per Corollary 1). $\square$

E.5 Proof of Corollary 4

From Theorem 1 and the observation that $\kappa \rightarrow \phi - 1$ and $u \rightarrow 1/(1 - \phi/\phi^2) = \phi/(\phi - 1)$ in the limit $\lambda \rightarrow 0^+$, we have $E_{test}(\hat{w}) \simeq \bar{E} + \zeta$, with

$$\begin{aligned} \bar{E} &= V + B, \quad B = r^2 \frac{(\phi - 1)^2}{\phi^2} \frac{1}{1 - 1/\phi} = r^2 (1 - 1/\phi), \quad V = \frac{\sigma^2}{\phi - 1}, \\ \zeta &= \frac{p_2 c^2}{\phi^2} \left(p_2 \left(1 + \frac{p_1}{\phi - 1} \right) + (p_1 + \phi - 1)^2 \right), \end{aligned}$$

and the first part of the result follows after some algebra.

The second part then follows from expanding the above around $\phi = \infty$. $\square$

F Proof of Proposition 3 and Theorem 2

F.1 Proof of Proposition 3

We state and proof a more general result without the requirement $\Sigma_1 = \Sigma_2 = \Sigma$.

Let (e_1, e_2, τ) be the unique nonnegative solution to the following fixed-point equations

$$e_1 = \frac{1}{1 + \psi \tau \text{tr} \Sigma_1 K^{-1}}, \quad e_2 = \frac{1}{1 + \psi \tau \text{tr} \Sigma_2 K^{-1}}, \quad \tau = \frac{1}{1 + \text{tr} K_0 K^{-1}}, \quad (114)$$

$$\text{with } K_0 := p_1 e_1 \Sigma_1 + p_2 e_2 \Sigma_2, \quad K := \gamma \tau K_0 + \lambda I_d. \quad (115)$$

Also, let (v_1, v_2, ω) to be the unique nonnegative solution to the following fixed-point equations

$$v_1 = \psi e_1^2 \text{tr} \Sigma_1 (\gamma \tau^2 L + \lambda \omega I_d) K^{-2}, \quad v_2 = \psi e_2^2 \text{tr} \Sigma_2 (\gamma \tau^2 L + \lambda \omega I_d) K^{-2}, \quad (116)$$

$$\omega = \tau^2 \text{tr} (\gamma K_0^2 + \lambda L) K^{-2}, \quad \text{with } L := p_1 v_1 \Sigma_1 + p_2 v_2 \Sigma_2 + \lambda B. \quad (117)$$

Finally, define $d \times d$ matrices C_1, C_2, D_1, D_2, E by

$$C_1 := \gamma p_1 e_1^2 (\gamma \tau^2 (B + p_2 u_2 \Sigma_2) + \omega I_d) \Sigma_1 + u_1 (\gamma \tau p_2 e_2 \Sigma_2 + \lambda I_d)^2, \quad (118)$$

$$C_2 := \gamma p_2 e_2^2 (\gamma \tau^2 (B + p_1 u_1 \Sigma_1) + \omega I_d) \Sigma_2 + u_2 (\gamma \tau p_1 e_1 \Sigma_1 + \lambda I_d)^2, \quad (119)$$

$$D_1 := \tau^2 e_1 B + (e_1 \omega - \tau v_1) I_d + \gamma \tau^2 p_2 (e_1 u_2 - e_2 u_1) \Sigma_2, \quad (120)$$

$$D_2 := \tau^2 e_2 B + (e_2 \omega - \tau v_2) I_d + \gamma \tau^2 p_1 (e_2 u_1 - e_1 u_2) \Sigma_1, \quad (121)$$

$$E := \gamma (\gamma \tau^2 B + \omega I_d), \quad (122)$$

Proposition 4. *In the proportionate scaling limit (6), it holds that*

$$r_j^{(1)}(A) \simeq \gamma \tau p_j e_j \text{tr} A \Sigma_j K^{-1}, \quad (123)$$

$$r_j^{(3)}(A, B) \simeq \gamma p_j A \Sigma_j C_j K^{-2}, \quad (124)$$

$$r_j^{(4)}(A, B) \simeq \gamma p_j \text{tr} A \Sigma_j D_j K^{-2}, \quad (125)$$

$$r^{(5)}(A, B) \simeq \text{tr} A E K^{-2}. \quad (126)$$